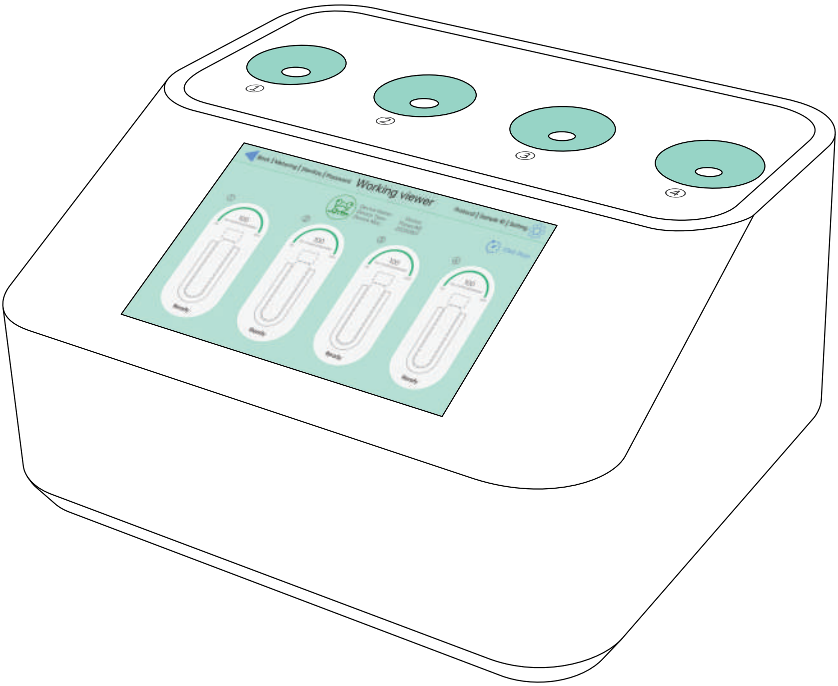
ThawLINE



User Manual

CONTENTS

Function description and appearance

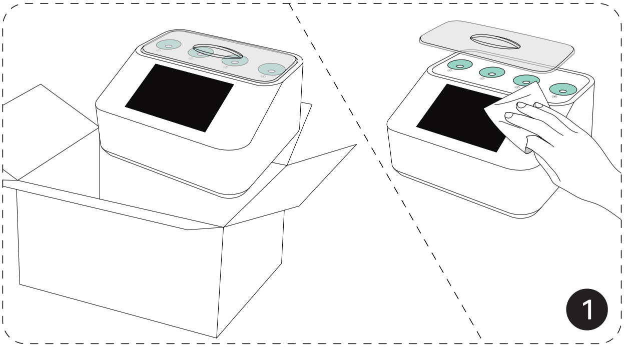
The frozen storage, recovery and freezing operation of samples have been paid more and more attention by researchers. As an indispensable step in biological experiments, their stability and contamination rate will directly affect the results of experiments. Therefore, a good set of products can better help researchers do experiments, and can get faster and better experimental results. Similarly, industrial users are also more in need of similar products to stabilize production and improve efficiency, Kemesser has focused on such research and development for ten years, launched a series of products can be a good solution to biological samples frozen storage, recovery, freezing operations and other processes, to provide customers with a better and more stable operating environment, cell recovery instrument for efficient thawing of a variety of specifications of containers of living cells. After the biological test tube is directly removed from liquid nitrogen or dry ice, it is inserted into the slot of the cell resuscitate instrument. The thawing system can quickly restore the biological vitality of the cells through sensor technology and temperature control technology to achieve the effect of recovery.

Suggested Settings

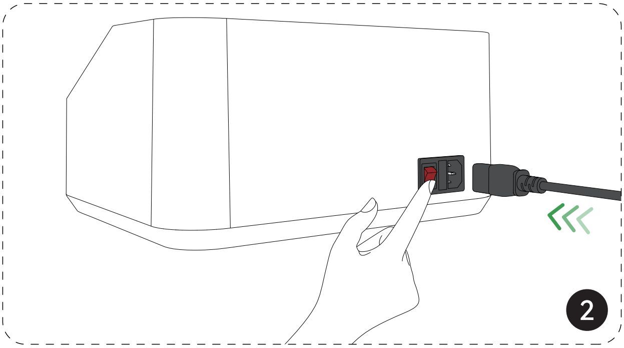
Note: Please set the parameters according to the volume of reagent in the container. The table below is for reference only.

Container \ Cryopreservation Environment	Ultra-low Temperature Freezer	Liquid Nitrogen
T2（2ml Cryogenic Tube)	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：60℃
T5（5ml Cryogenic Tube)	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：60℃
A1（1ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：60℃
A2（2ml Cryogeni Storage Vial)	Ice Crystal Parameter Settings：70~80 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：70~80 Preheating Temperature Settings：60℃
A6（6ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings：70~80 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：70~80 Preheating Temperature Settings：60℃
A10（10ml Cryogenic Storage Vial)	Ice Crystal Parameter Settings：120~140 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：120~140 Preheating Temperature Settings：60℃
C2（2ml Glass Vial)	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：40~70 Preheating Temperature Settings：60℃
C6（6ml Glass Vial)	Ice Crystal Parameter Settings：70~80 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：70~80 Preheating Temperature Settings：60℃
C10（10ml Glass Vial)	Ice Crystal Parameter Settings：140~160 Preheating Temperature Settings：58℃	Ice Crystal Parameter Settings：140~160 Preheating Temperature Settings：60℃

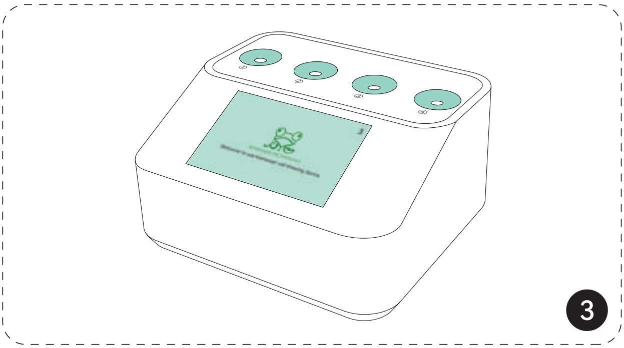
Quick Start Guide



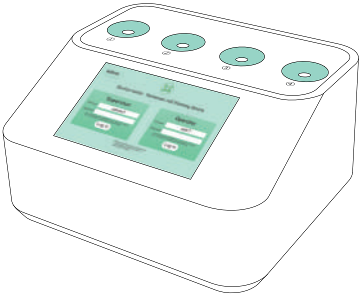
Remove the instrument from the carton, take off the dust cover, and wipe the surface of the device.



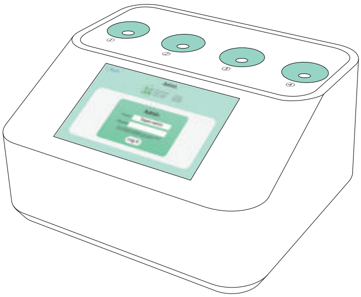
Carefully connect the power cord from the accessories to the power socket on the back of the device, then turn on the power switch.



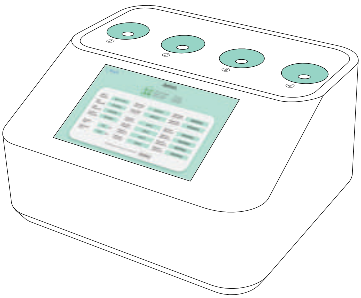
Enter the welcome screen.



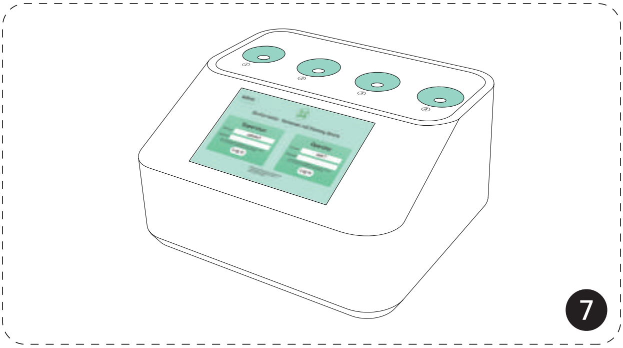
Click【Administrator】at the top left of the login interface.



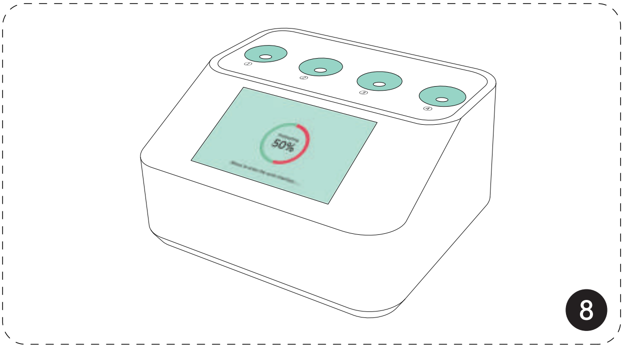
Enter the password as Admin.



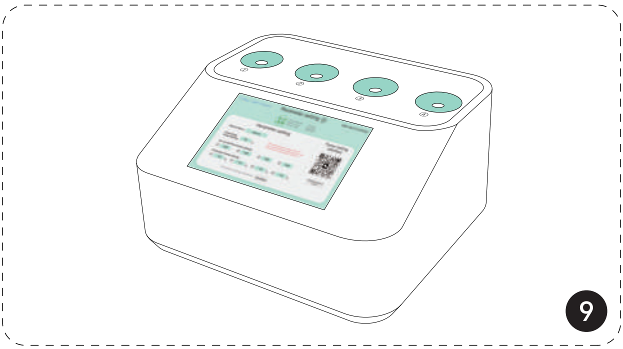
Enter the【Administrator】interface to assign accounts and set passwords. After completing the settings, click Confirm to save the data.



Enter the password as Supervisor.

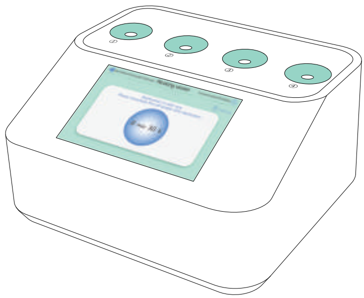


Wait for preheating.



After preheating is complete and entering the working interface, click the top right corner to enter “Settings”. Adjust the Ice Crystal Parameters and other parameters, then click Confirm to save the data.

Note: Ice crystal parameters are a critical factor in the thawing process. Since each user's sample volume and cryopreservation solution may vary, we strongly recommend performing trial thawing with the same type and volume of cryopreservation solution before actual use to determine the optimal ice crystal settings. The ideal thawing state is achieved when 5%–15% of ice crystals remain at the end of the process.



Return to the working interface and click 【Sterilize】 at the top left to enter the sterilization interface. Wait until sterilization is complete.

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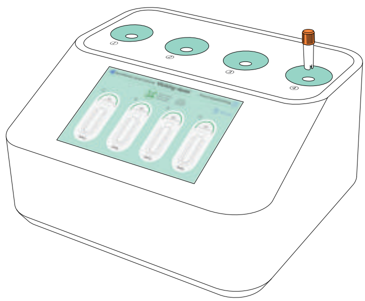

Return to the working interface and insert the sample container to start cell thawing.

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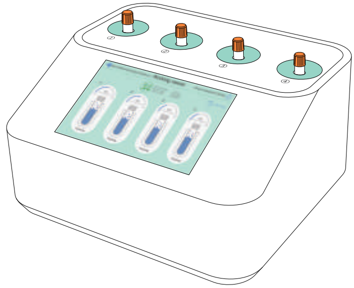

Enter the thawing interface, the cells are being thawed.

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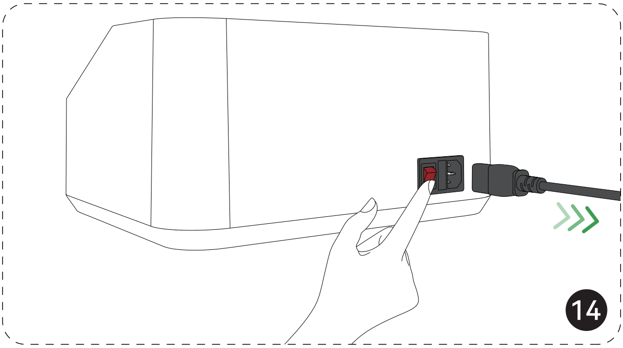
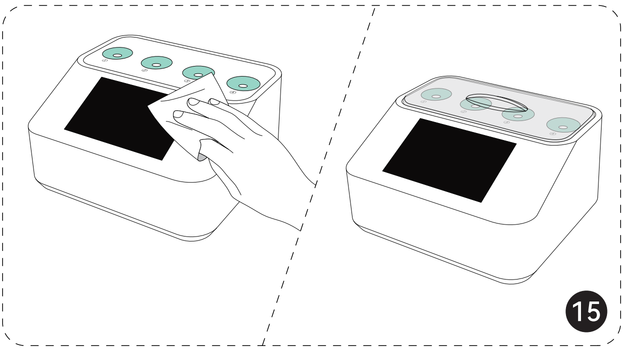
After cell thawing is complete, promptly remove the sample container.

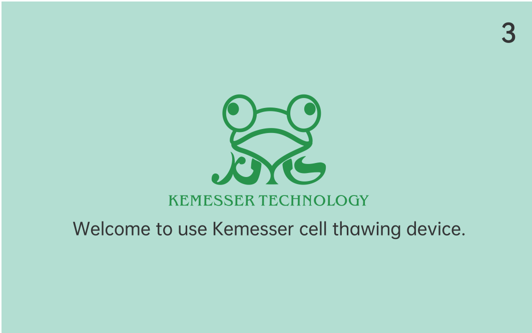
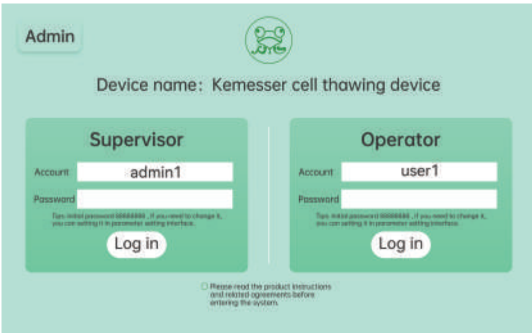
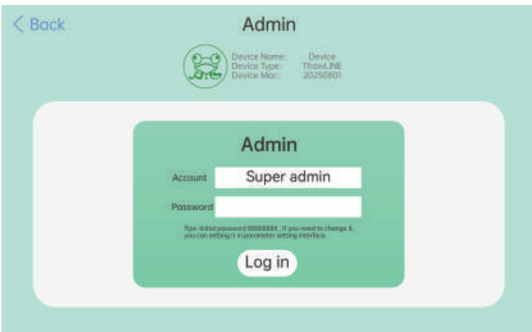


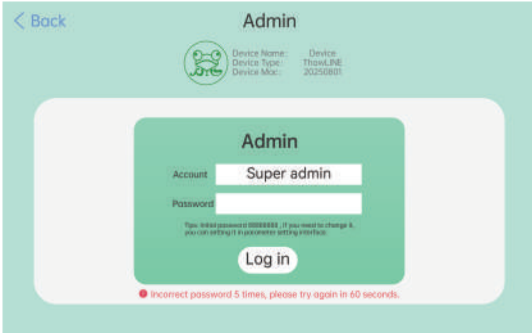
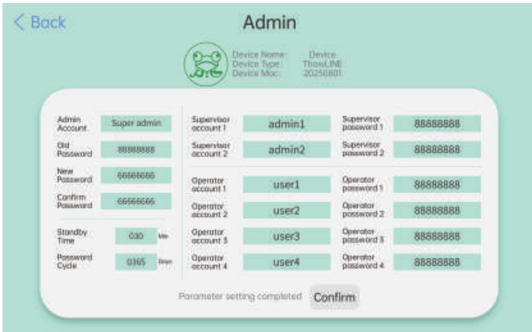
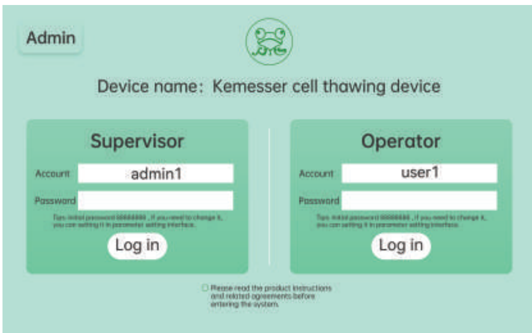
After placing the sample container, turn off the power switch at the back of the device and carefully unplug the power cord.

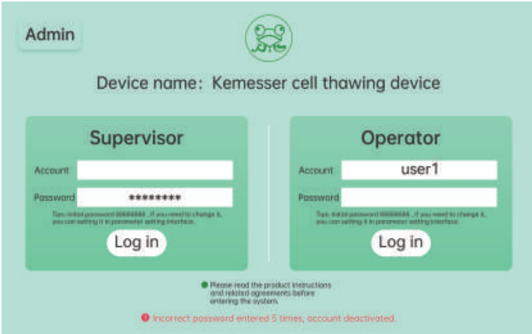
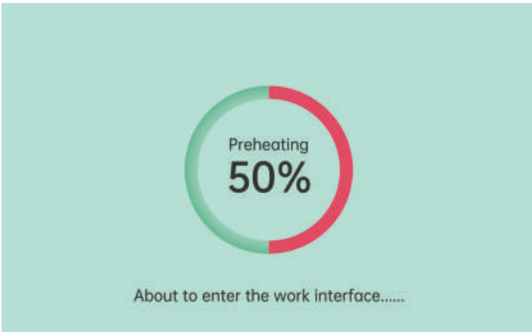
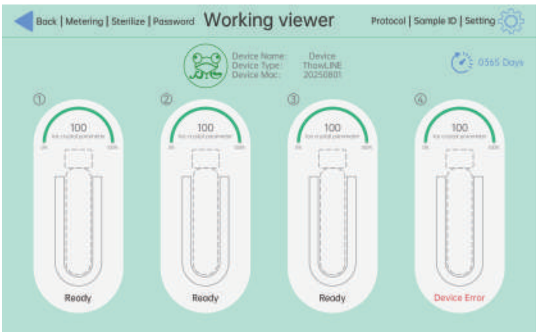


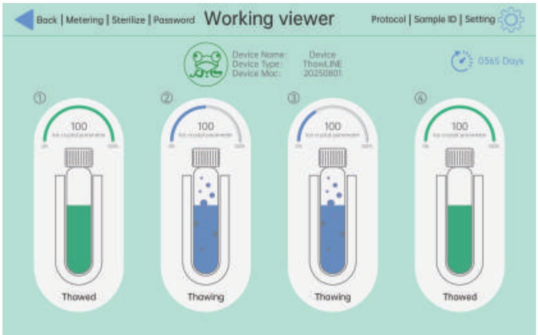
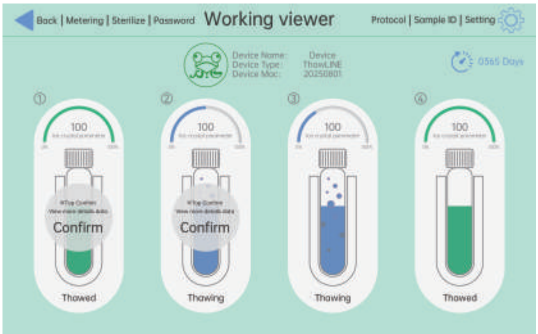
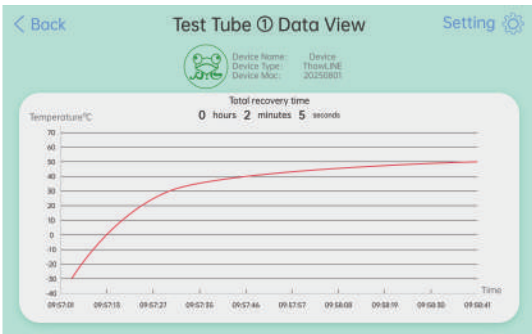
After wiping the device surface, replace the dust cover.

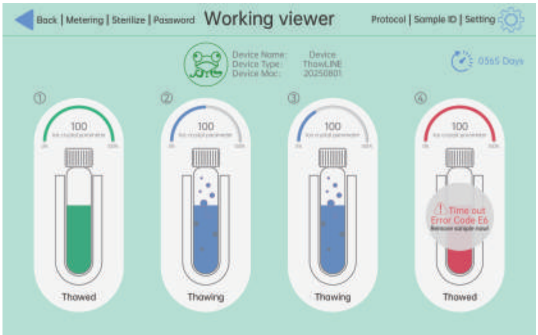
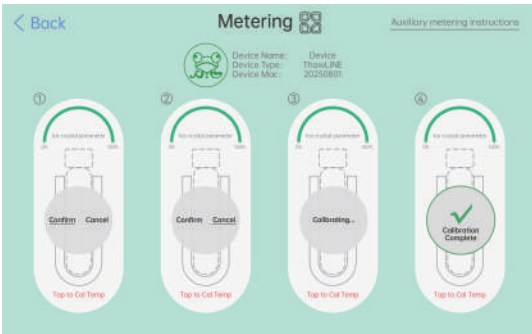
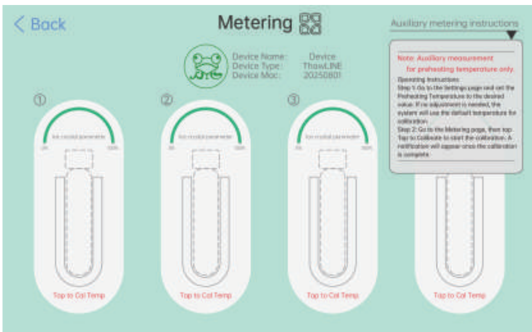
Software Interface

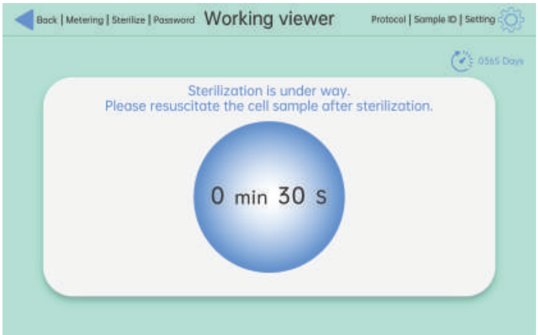
No.	Steps	Display Screen Indication	Sound
1	Check that the instrument is in good condition and connect the power supply.	N/A	N/A
2	**Welcome Screen** With normal power supply, toggle the switch on the back of the instrument. The display screen will light up automatically, as shown on the right.		N/A
3	**Login Screen** Click [Admini] at the top left of the login screen to enter the administrator interface.		N/A
4	**Admini Interface** The device's initial administrator account is **Super admin,** with the password **88888888.**		N/A

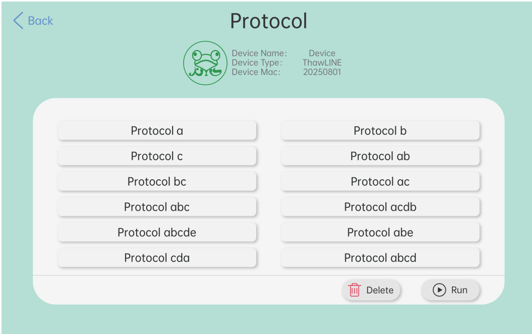
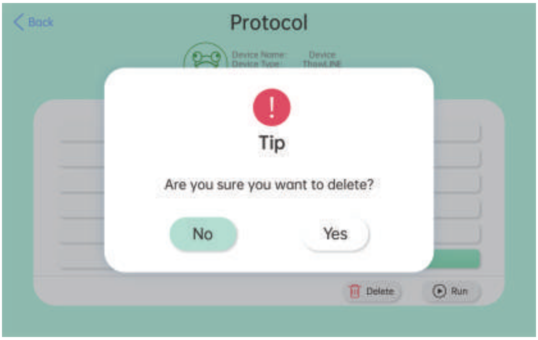
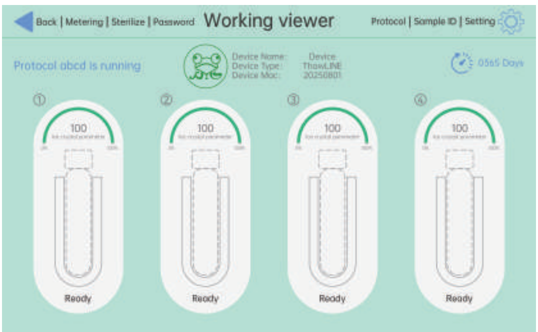
No.	Steps	Display Screen Indication	Sound
5	**Incorrect Password** If the password is entered incorrectly 5 times, you must wait 60 seconds before trying again.		N/A
6	**Admini Interface** In the administrator interface, you can set the administrator account and password, standby time (if there is no operation within the set time, the device will automatically return to the login screen), password cycle (supervisor and operator passwords have a time limit; if not updated before expiration, the account is locked and the login screen shows a password error requiring higher-level permission to unlock), as well as supervisor and operator accounts and passwords. After modifying this interface, you must enter the admini's old password before clicking Confirm to save the changes.		N/A
7	**Login Screen** Return to the login screen and enter the supervisor account and password.		N/A

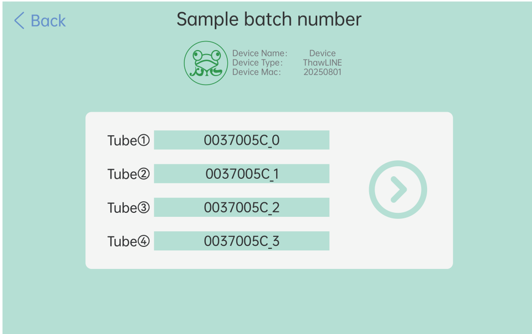
No.	Steps	Display Screen Indication	Sound
8	**Incorrect Password** If the password is entered incorrectly 5 times, the device will log out the current account (data within the account will be retained).		N/A
9	**Preheating Interface** After a successful login, the device begins preheating.		N/A
10	**Working Interface** The interface contains four sample container icons labeled ①/②/③/④. The Ice Crystal Parameter indicates the ice crystal size after thawing (the larger the value, the smaller the ice crystals). If a device fault occurs, [Device Error] will be displayed below the sample container icon. Click [Return] to go back to the login interface and switch accounts.		N/A

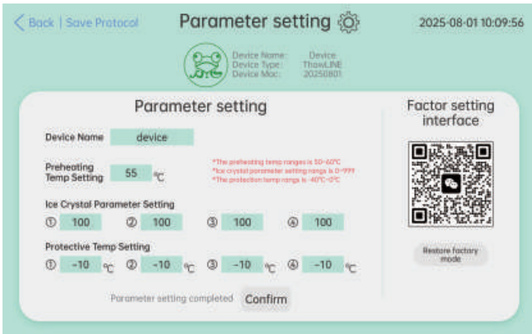
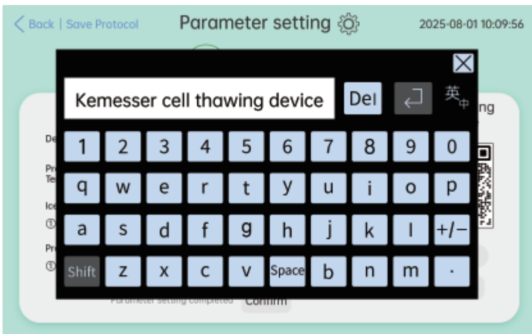
No.	Steps	Display Screen Indication	Sound
11	**Working Interface** Insert the sample container into the device slot. "Thawing" is blue, and "Thawed" is green, as shown in the figure on the right.		N/A
12	**View Tube Data** Click the sample container icon to pop up the View Tube Details button. Click Confirm to enter the tube data dashboard.		N/A
13	**Data View** The interface, as shown on the right, displays the detailed data curves of the sample.		N/A

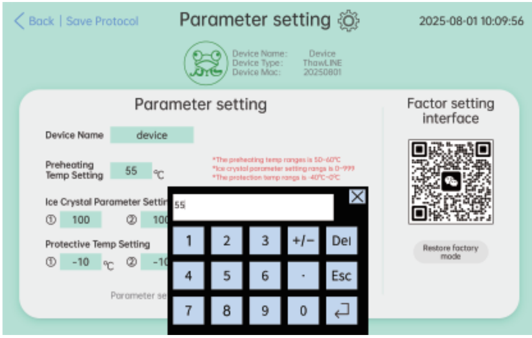
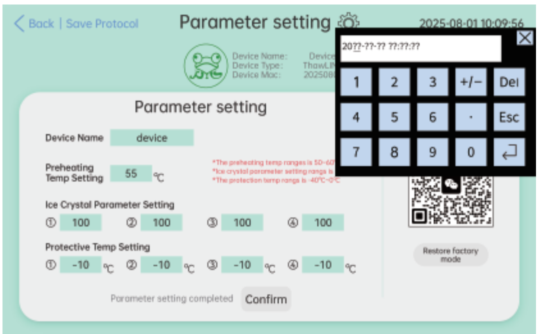
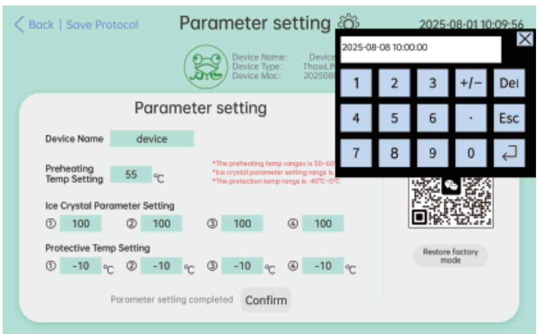
No.	Steps	Display Screen Indication	Sound
14	**Working Interface** If the sample is not removed within 1 minute after thawing is complete, the device will trigger an alarm. The container icon turns red, as shown in the figure on the right.		N/A
15	**Metering Interface** The main functions of the measurement interface: • Calibrate the set heating element parameters. • Measurement: After calibration, use a standard instrument to check the accuracy of the device (e.g., if the heating element is set to 55°C, the standard instrument can verify whether the heating element temperature is accurate).		N/A
16	**Metering Interface** Click Metering to display the metering instructions.		N/A

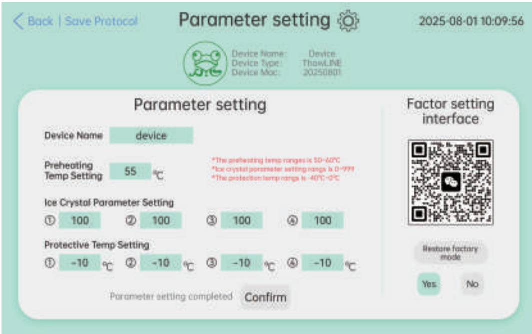
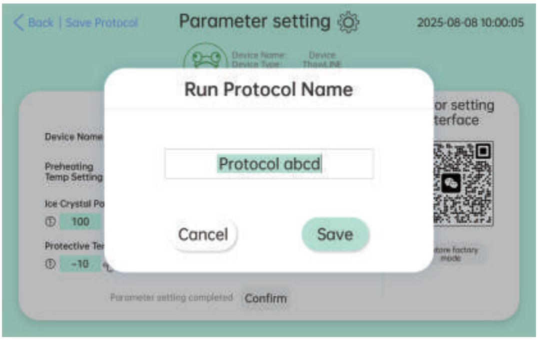
<table>
<tr><th>No.</th><th>Steps</th><th>Display Screen Indication</th><th>Sound</th></tr>
<tr><td>17</td><td>Sterilization Interface
Click [Sterilize] to start the device's ozone function with a 30-second countdown. After sterilization is complete, the interface returns to the main work screen. The interface cannot be operated during sterilization unless the device is powered off and restarted.</td><td>
</td><td>N/A</td></tr>
<tr><td>18</td><td>Password Service
Enter Password Service as a supervisor to set supervisor and operator passwords.
After modifying this interface, you must enter the supervisor's old password before clicking Confirm to save the changes.</td><td>
</td><td>N/A</td></tr>
<tr><td>19</td><td>Password Service
Enter Password Service as an operator to set the operator password.
After modifying this interface, you must enter the operator's old password before clicking Confirm to save the changes.</td><td>
</td><td>N/A</td></tr>
</table>

No.	Steps	Display Screen Indication	Sound
20	**Run Protocol** In the Settings interface, click [Save Run Protocol] at the top left to package the configured parameters into a running plan and save it to this interface. Supervisors can enable or delete running plans, while operators can only enable running plans.		N/A
21	**Run Protocol** When deleting a run protocol, a confirmation prompt window will appear.		N/A
22	**Run Protocol** After the run protocol is successfully enabled, the top left of the working interface will display that the protocol is running.		N/A

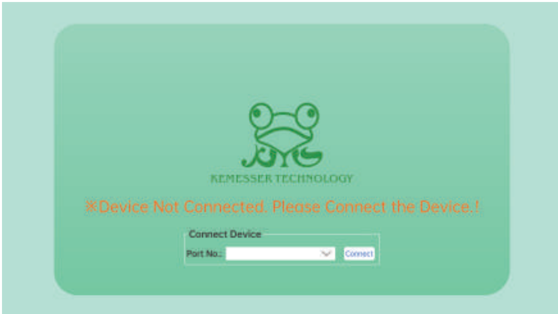
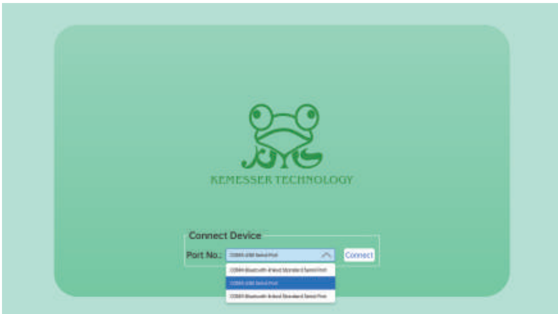
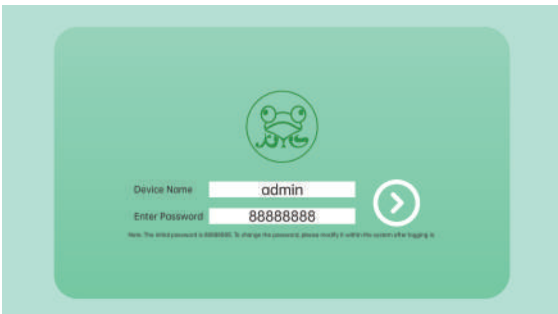
<table>
<tr><th>No.</th><th>Steps</th><th>Display Screen Indication</th><th>Sound</th></tr>
<tr><td>23</td><td>Sample Batch No.
The sample batch number for each container can be set individually. After entering the batch number, click the arrow button to save the sample batch number.</td><td>
</td><td>N/A</td></tr>
<tr><td>24</td><td>Settings Interface
You can set:
• Device Name
• Preheating Temperature
• Ice Crystal Parameters
• Protection Temperature
• Restore Factory Settings

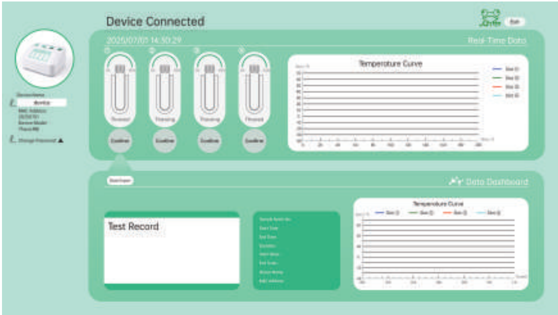
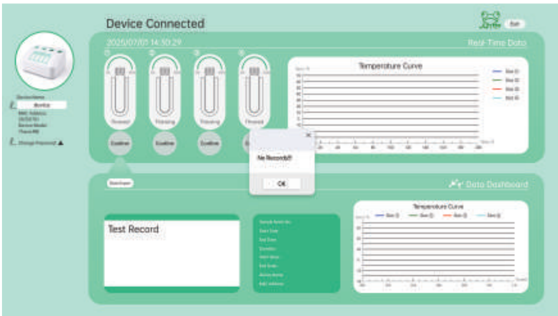
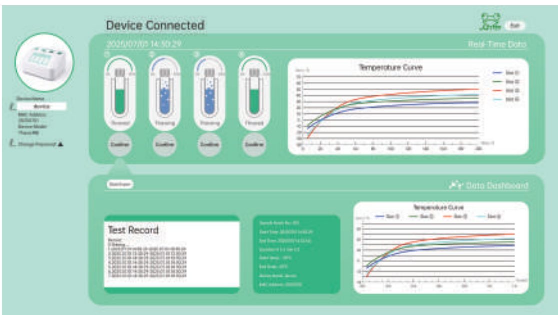
You can save run protocols and set the date and time (press and hold the date/time at the top right to set).</td><td>
</td><td>N/A</td></tr>
<tr><td>25</td><td>Name & Account Modification
• Device model and device MAC address cannot be changed.
After completing all settings on the interface, click Confirm to save the changes.</td><td>
</td><td>N/A</td></tr>
</table>

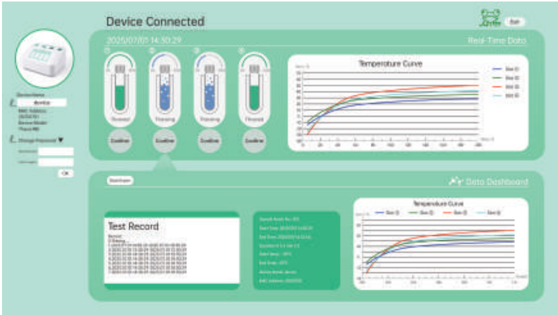
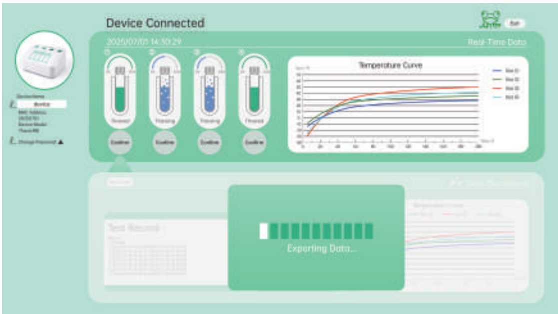
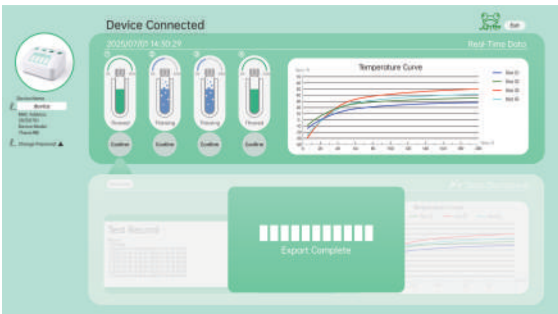
No.	Steps	Display Screen Indication	Sound
26	**Preheating Temperature, Ice Crystal Parameters, Protection Temperature** Click the input fields for Preheating Temperature, Ice Crystal Parameters, and Protection Temperature in Settings to modify them. After completing all settings, click Confirm to save the changes.		N/A
27	**Date & Time Settings** Press and hold the date and time at the top right of Settings to bring up the input keypad. Enter the desired date and time, then press Enter to save the changes.	 	N/A

No.	Steps	Display Screen Indication	Sound
28	**Restore Factory Settings** Click 【Settings】, then click [Restore Factory Settings] at the bottom right. A prompt [Yes] / [No] will appear, select as needed.		N/A
29	**Save Run Protocol** Click [Save Run Protocol] at the top left of the 【Settings】 interface to open the run protocol name input box.		N/A
30	**Save Run Protocol** If the run protocol name already exists, you need to re-enter a new name. The interface displays [Protocol name is incorrect].		N/A

PC Software Interface

No.	Steps	Display Screen Indication	Sound
1	Check that the device is in good condition and connect the power supply.	N/A	N/A
2	**Connect Device** After starting the software, if the interface displays "Device Not Connected. Please Connect the Device!", please check the following: • Connect the cell thawing device to the computer via the data cable. • Once the system detects the connection, the prompt will disappear automatically.		N/A
3	**Connect Port** After a successful connection, select the corresponding port number and click Connect to enter the login interface.		N/A
4	**Login to System** Enter the password on the login interface and click the arrow on the right to log in. • If the password is incorrect, a login failure message will appear. • After successful login, you will enter the main interface.		N/A

No.	Steps	Display Screen Indication	Sound
5	**Enter Main Interface** The main interface shows the device status as "Device Connected", with four slots displayed below. • All slots currently show No Data, indicating that no experiment has been run or there is no historical data. • You can click any slot to view details.		N/A
6	**No Records** At this point, no experiment has started. Clicking Data Export will display No Records.		N/A
7	**View Slot 1 Details** After starting the experiment, click the first slot to enter the detailed information page, which includes the following modules: • Test Records: Displays all experiment records for this slot. • Device Information: Shows the module's sample batch number, device name, etc. • Experiment Data: Displays the data collected during the slot's operation. • Temperature Curve: Shows the temperature changes throughout the experiment.		N/A

No.	Steps	Display Screen Indication	Sound
8	Click the second slot; the content is the same as Slot 1, and other slots operate similarly. Click Change Password to open fields for New Password and Confirm Again, allowing you to modify the password.		N/A
9	**Data Export** After initiating the export, the system will display "Data Export in Progress". Do not close the program or disconnect the device during this process.		N/A
10	**Export Complete** After the export is finished, the system will display "Export Complete". • The file will be saved to the default path.		N/A

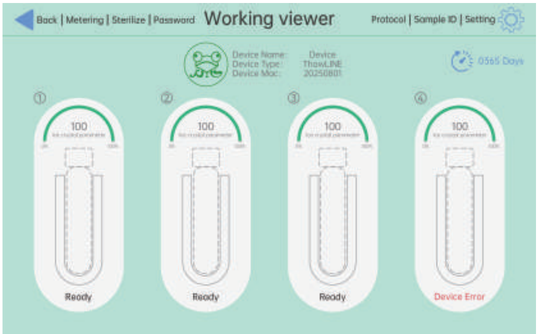
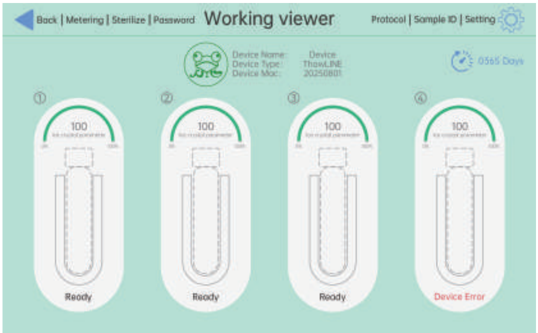
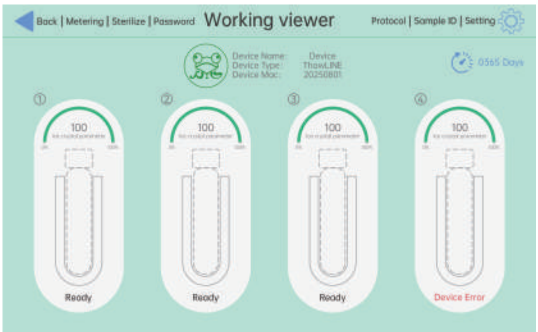
Parameter Settings and Function Usage

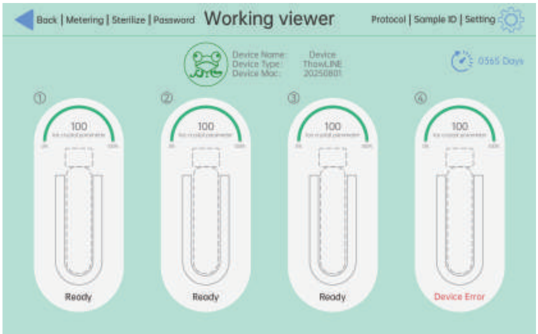
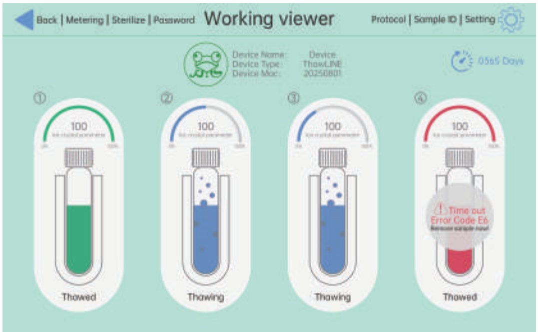
Parameter Settings and Function Usage			
Function Items	**Admini**	**Supervisor**	**Operator**
Assign Supervisor Account	√	×	×
Assign Operator Account	√	×	×
Delete (Revoke) Account	√	×	×
Change Admini Password	√	×	×
Change Supervisor Password	√	√	×
Change Operator Password	√	√	√
Ice Crystal Parameter Settings	×	√	×
Preheating Temperature Settings	×	√	×
Protection Temperature Settings	×	√	×
Modify Device Name	×	√	×
Restore Factory Settings	×	√	×
Modify Date & Time	×	√	×
Measurement Calibration	×	√	×
One-Click Sterilization	×	√	√
Sample Batch No.	×	√	√
View Tube Data Dashboard	×	√	√
Cell Thawing Operation	×	√	√
Bluetooth Printing	×	√	√
Enable Run Protocol	×	√	√
Save and Delete Run Protocol	×	√	×
Automatically Save Thawing Data to SD Card	×	√	√
Real-Time Temperature Curve Monitoring	×	√	√

Note:

1. Administrator can only access the Administrator Interface, cannot perform operations or modify core parameters.
2. Supervisor has full operational permissions, except for the Administrator Interface, which cannot be modified.
3. Operator can only perform Sample Batch No. Setting, One-Click Sterilization, Sample Batch No., Real-Time Temperature Curve Monitoring, Automatically Save Thawing Data to SD Card, and Password Service Interface Settings, cannot modify core parameters.

Alarm List

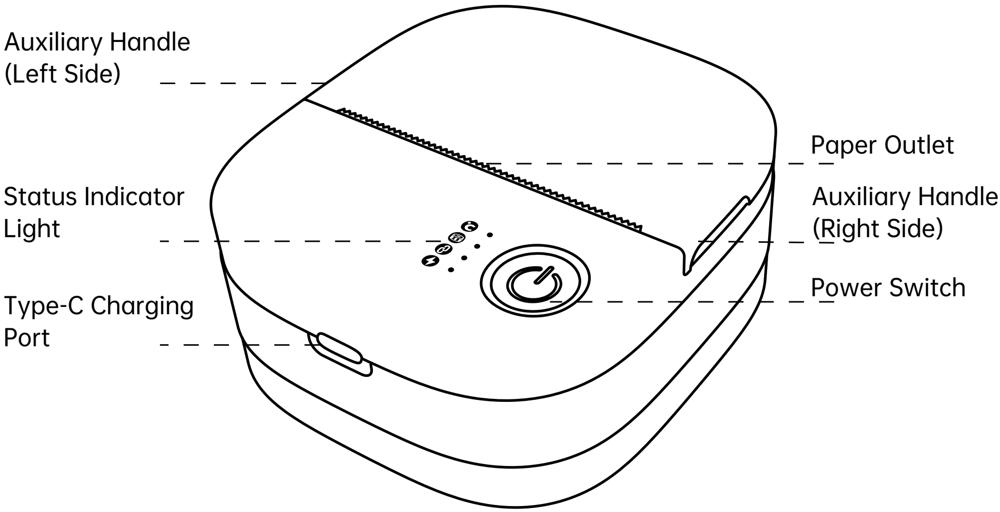
<table>
<tr><th>No.</th><th>Status & Solutions</th><th>Display or Sound Alerts</th></tr>
<tr><td>1</td><td>If "Device Error" Appears Below the Tube Icon on the Display or a Sound Alert Occurs.
The infrared sensor is abnormal. Please perform a Factory Reset.
If the error persists after restarting, contact Kemesser technical support.</td><td>
</td></tr>
<tr><td>2</td><td>If "Device Error" Appears Below the Tube Icon on the Display.
NTC temperature detection is abnormal (temperature reads zero, too high, or too low).
Please contact Kemesser technical support.</td><td>
</td></tr>
<tr><td>3</td><td>If "Device Error" Appears Below the Tube Icon on the Display.
The heating element current is either too high or zero. Please contact Kemesser technical support to replace the heating unit.</td><td>
</td></tr>
</table>

No.	Status & Solutions	Display or Sound Alerts
4	If "Device Error" Appears Below the Tube Icon on the Display. The electromagnet current is either too high or zero, preventing the heating unit from being held. Please contact Kemesser technical support.	
5	If the Display Shows "Timeout – Please Remove Promptly". If the sample is not removed within 1 minute after thawing is complete, the device will trigger an alarm, and the tube icon will turn red.	
6	If the Device Slot Cannot Hold the Sample Container. The sample temperature is too high. Remove the high-temperature sample container and insert a low-temperature sample container to retry. If the alarm persists, contact Kemesser technical support.	If the Device Emits a "Beep" Sound.

Bluetooth Printer Operation Instructions

This device is equipped with a Bluetooth thermal printer for automatically printing cell thawing process reports and preserving critical experimental data.

1. Product Overview



Print Head Temperature
(When the print head temperature is within the normal range, the corresponding indicator remains steadily lit. If the indicator flashes, wait until it turns off and the temperature decreases before continuing use.)

Paper-Out Detection
(The indicator remains steadily lit when paper is loaded in the paper compartment, it flashes when the printer is out of paper.)

Bluetooth Connection
(The indicator remains steadily lit when the Bluetooth connection is successful, it flashes when the connection fails.)

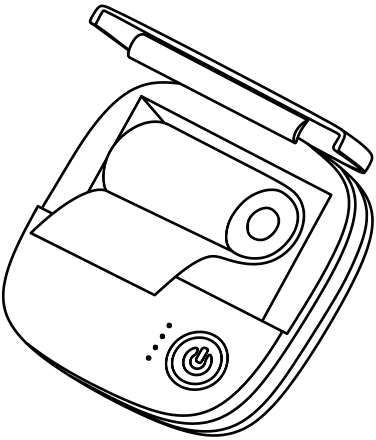
Power Supply
(The indicator remains steadily lit when the device is powered on and operating normally.)

2. Device Preparation

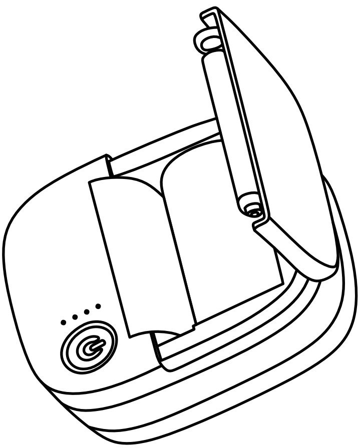
Power Supply: Fully charge the printer or connect it to a USB power source via the Type-C port.
Power On: Press and hold the printer power button to turn on the device. The printer will automatically enter Bluetooth pairing mode.

3. Thermal Paper Installation and Replacement

When the thermal paper is depleted, print quality decreases, or a paper jam occurs, follow the steps below to replace the compatible thermal paper:

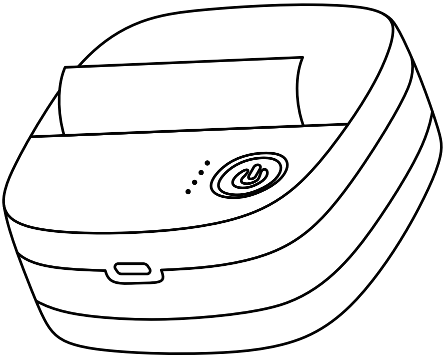


① Open the paper compartment: Pinch the auxiliary handles on both sides of the printer and lift upward to open the top cover.



② Replace the paper roll: Remove the used paper roll and place a compatible thermal paper roll into the paper compartment. Ensure the following requirements are met:

- The smooth printable side of the paper end must face downward.
- The paper feed direction of the roll must face the paper outlet at the front of the device.
- The paper roll must be properly installed without misalignment or jamming.



③ Paper Path Calibration: Pull approximately 2 cm of the paper end outward, then close the top cover. The printer will automatically feed and calibrate the paper. Printing can begin once calibration is completed.

Precautions:

● Please use thermal paper compatible with this device. Low-quality or incompatible paper may cause blurred printing, paper jams, or damage to the print head.

● If the paper is installed in the wrong direction, printing will not function properly and the paper orientation must be adjusted.

● Avoid touching the print head when replacing paper to prevent burns or contamination.

4. Bluetooth Connection and Automatic Printing Process

① After the cell thawing instrument is powered on, it will automatically search for and connect to the enabled Bluetooth printer.

② When the Bluetooth connection status indicator on the printer (the third indicator light) remains lit, the connection with the cell thawing instrument has been successfully established.

③ When the cell thawing process for a single chamber is completed and the sample container is automatically ejected, the printer will automatically start printing and generate a complete report for the thawing session.

5. Print Report Description

The automatically generated thawing report includes the following key experimental parameters:

● StartTime (Thawing start time)

● FinishTime (Thawing completion time)

● ThawTime (Total thawing duration)

● InputTemp (Sample input temperature)

● LeaveTemp (Sample output temperature)

● Port (Chamber number)

● Sample Number (Sample batch number)

● Device (Device name)

● User (Operator)

● MAC (Instrument MAC address)

6. Routine Maintenance and Troubleshooting

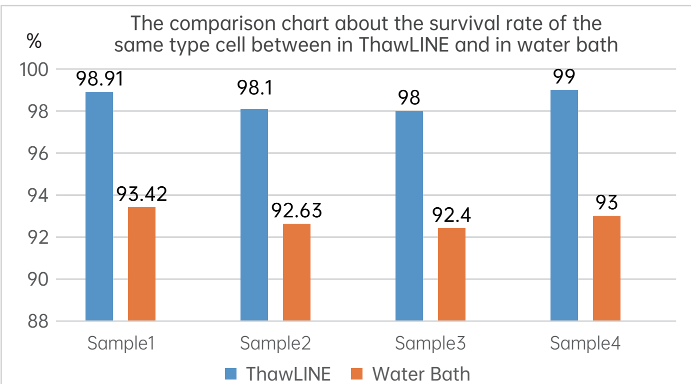
Issue	Troubleshooting and Solution
Bluetooth indicator is off and connection cannot be established	1. Check whether the printer is powered on and has sufficient battery power. 2. Restart the printer and the cell thawing instrument, then reconnect them. 3. Ensure that the printer's Bluetooth connection is not occupied by other devices.
No automatic printing after thawing is completed	1. Check whether the printer is out of paper or the paper is installed in the wrong direction. 2. Clean any dirt or residue from the surface of the print head. 3. Confirm that the Bluetooth connection status is functioning properly.
Printed content is blurred / incomplete	1. Replace with compatible thermal paper. 2. Gently clean the print head with a clean cotton swab to remove dust or stains. 3. Check whether the thermal paper is damp or expired.
Paper jam / abnormal paper feeding	1. Open the paper compartment and remove the jammed paper. 2. Check whether the paper roll is properly installed and recalibrate the paper path. 3. Avoid using damp or wrinkled thermal paper.

Package Contents and Technical Specifications

List	Quantity	Notes
Cell Thawing Device	1	Device
Power Cord	1	220V
SD Card	1	
Card Reader	1	
Dust Cover	1	
Temperature Probe	1	
User Manual	1	

Performance Parameters

Thawing Time	2–8 minutes (varies depending on the cryobag sample)
Rated Power	Maximum 300W
Input/Output Voltage (Power Adapter)	AC 100–240V / DC 12V, 50–60Hz
Output Current	15A
Power Plug	US,UK,EU
Dimensions	L × W × H: 400mmX445mmX255mm
Certification	CE
Thawing Method	Waterless Dry Heat Transfer
Initial Temperature Protection	Yes, users can set it themselves.
Temp. Calibration	Yes, with intelligent calibration for easy third-party measurement.
Sterilize	Yes, the instrument automatically sterilizes from the inside out.
Ice Crystal Control	Customers can adjust the size of ice crystals after resuscitation according to their actual usage needs.
Permission Management	Three Levels: Administrator / Supervisor / Operator
Thawing Complete	With Audible Alert
Display Screen	Touchscreen display, allows real-time monitoring of the thawing chamber.
Data Export	Data is stored on the SD card and can be exported via the host computer.
Online Printing	Equipped with a printer, supports online printing, and can connect directly to the printer.
Permission Levels	Date and time functions are locked, only supervisors can modify them.



Safety Instructions

The automated cell thawing system is designed for use by well-trained and experienced laboratory personnel.

1. Please read the instructions before installing, using or cleaning the equipment;
2. To prevent risk of severe or fatal electric shock, use only the AC power supply provided with this package;
3. Do not place near or immerse in water or other liquids;
4. Unauthorized removal of the casing will void your valid warranty period. Do not remove the casing of the device unless authorized by Kemesser;
5. There is a jack on the back of the machine. Do not insert any plug into this port or attempt to connect the device to other devices (such as DC power supply, computer) unless authorized by Kemesser;
6. This device is designed for indoor benchtop laboratories. Make sure the device is safely placed on a level surface during use. It is not recommended to use it near strong vibrations such as centrifuges, shakers or vortex mixers;
7. Make sure the tubes are frozen in an upright and vertical position so that the frozen contents are at the bottom of the tube;
8. Make sure the cryogenic tubes have a clean appearance and that the caps are tightly sealed to prevent leaks;
9. Insert the cryogenic tube into the opening of the device and start the thawing process. Do not touch the tube with your hands or any other objects until the thawing process is completed;
10. The thawing time is generally 2-3 minutes. During use, laboratory personnel recommend staying close to the instrument so that it can be transferred immediately after thawing is completed;
11. Keep the device covered when not in use;
12. When the device is turned on, the heating element inside the vial port is heated. Avoid inserting body parts (such as fingers or any object other than the cryogenic vial) into thethawing port at any time.

Care and Cleaning

Do not attempt to clean the internal components. Do not insert any cleaning objects or solutions into the open thawing block on the top of the machine. Before or after each use, wipe the exterior with a soft, non-abrasive lab wipe moistened with 70% alcohol. Thawing condensation does not cause problems as it evaporates during the defrosting process or during standby.

Technical Service and Repair

Contact Kemesser technical support at 0571-87156759 or email andy@kemesser.com for help.

Certificate of Conformity:

Certificate of Conformity	
Name:	Automated cell thawing instrument
Model:	
Date:	
Test Result:	Qualified
Inspection date:	

Warranty Card:

Warranty:

1.Enjoy ONE year free warranty for the whole machine from the date of purchase;

2.The following situations are not covered by the warranty:

A: Failure and damage caused by not operating correctly according to the manual;

B: Failures and damages caused by self-disassembly and repair;

C: Failures and damages caused by human factors;

D: If the warranty card cannot be presented or the instrument does not have the official sales seal.

Warranty				
No.	Fault cause	Repair dare	Repair man	Remark
1				
2				
3				
4				
5				
6				
7				

KEMESSER
TECHNOLOGY

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